

Solutions for Analysis I by Terence Tao

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Remark. This document follows the third edition of *Analysis I*.

1 Introduction

No exercises from this chapter.

2 Starting at the beginning: the natural numbers

2.1 The Peano axioms

No exercises from this subchapter.

2.2 Addition

Exercise 2.2.1. *Prove Proposition 2.2.5 (Associativity of addition): For any natural numbers a, b, c , we have $(a + b) + c = a + (b + c)$.*

Solution. We induct on a , keeping b and c fixed.

First, consider the base case $a = 0$. Then we have $(0 + b) + c = 0 + (b + c)$. By Lemma 2.2.2, we obtain $b + c = b + c$ as desired.

Now we assume inductively that $(a + b) + c = a + (b + c)$; we now have to show $((a++) + b) + c = (a++) + (b + c)$ to close the induction. By applying Lemma 2.2.3 twice on the left-hand side, $((a++) + b) + c = ((a + b)++) + c = (a + b + c)++$. Similarly, applying Lemma 2.2.3 on the right-hand side yields $(a++) + (b + c) = (a + b + c)++$. Thus, both sides are equal, and we have closed the induction. $\square$

Exercise 2.2.2. *Prove Lemma 2.2.10: Let a be a positive number. Then there exists exactly one natural number b such that $b++ = a$.*

Solution. We induct on a .

First, consider the base case $a = 1$. By Definition 2.1.3, $1 = 0++$. Suppose there is another natural number b such that $b++ = 1$. However, by Axiom 2.4, if $b++ = 0++ = 1$, then $b = 0$. Thus, there is exactly one natural number b such that $b++ = 1$. Hence, the base case is proven.

Suppose inductively that there exists exactly one natural number b such that $b++ = a$. We now have to show there exists exactly one natural number b' such that $b'++ = a++$. By the inductive hypothesis, $a++ = (b++)++ = b'++$. As per Axiom 2.4, there exists exactly one solution for b' , namely $b' = a$. This closes the induction. $\square$

Exercise 2.2.3. *Prove Proposition 2.2.12 (Basic properties of order of natural*

numbers).

Solution. Let a, b, c be natural numbers. Then

(a) *Reflexivity:* $a \geq a$.

By Lemma 2.2.2, $a + 0 = a$. Thus, by Definition 2.2.11, $a \geq a$.

(b) *Transitivity:* If $a \geq b$ and $b \geq c$, then $a \geq c$.

By Definition 2.2.11, write $a = b + m$, $b = c + n$, where m and n are natural numbers. Thus, $a = (c + n) + m = c + (n + m)$ by associativity of addition. By mathematical induction, it is trivial to prove that $n + m$ is a natural number. Therefore, $a \geq c$. $\square$

(c) *Anti-symmetry:* If $a \geq b$ and $b \geq a$, then $a = b$.

By Definition 2.2.11, if $a \geq b$, then for a natural number m , $a = b + m$. But also if $b \geq a$, then $b = a + n$, where n is a natural number. Thus $a = (a + n) + m = a + (n + m)$. By Lemma 2.2.2 and Proposition 2.2.6 (cancellation law), $0 = n + m$. By Corollary 2.2.9, this results in $m = n = 0$. Thus $a = b + 0 = b$ as required. $\square$

(d) *Preservation of order:* $a \geq b$ if and only if $a + c \geq b + c$.

First, prove $a \geq b \implies a + c \geq b + c$. According to Definition 2.2.11, if $a \geq b$, then $a = b + m$, where m is a natural number. Thus $a + c = (b + m) + c = (b + c) + m$, thus $a + c \geq b + c$.

Conversely, consider $a + c = (b + c) + m$, where m is a natural number. By Proposition 2.2.6 (cancellation law) and the associativity of addition, $a = b + m$, meaning $a \geq b$.

Because $a \geq b \implies a + c \geq b + c$ and $a + c \geq b + c \implies a \geq b$, $a + c \geq b + c \iff a \geq b$. $\square$

(e) $a < b$ if and only if $a++ < b$.

First, consider $a < b \implies a++ \leq b$. According to Definition 2.2.11, if $a < b$, then $b = a + m$, where m is a natural number. Note that $m \neq 0$ because we would otherwise obtain $a = b$, which is contradictory to the given condition $a < b$. Therefore, by Axiom 2.3, we have $m = n++$, where n is a natural number. Thus, by the definition and commutativity of addition, $b = a + m++ = (a + m)++ = (a++) + m$. It follows that $a \leq b$.

Now consider the converse. From $a++ \leq b$, we obtain $b = a++ + p$, where p is a natural number. Thus, by the definition and commutativity of addition, $b = a++ + p = (a + p)++ = a + p++$. According to Axiom 2.3, 0 is not the successor of any natural number, so $p++ \neq 0$. Hence, $a \leq b$.

Because $a < b \implies a++ \leq b$ and $a++ \leq b \implies a < b$, $a < b \iff a++ \leq b$. $\square$

(f) $a < b$ if and only if $b = a + d$ for some positive number d .

First consider $a < b \implies b = a + d$. From $a < b$ we know $a \geq b$, or $b = a + d$ for a natural number d . For the sake of contradiction, suppose $d = 0$. Therefore $a = b$, which contradicts $a < b$. Hence, d is positive; thus, $a < b \implies b = a + d$ for some positive number d .

Now consider the converse. From Definition 2.2.11, $b = a + d$ implies $b \geq a$, or $a \leq b$. If $b = a$, by Proposition 2.2.6 (cancellation law), $d = 0$, which contradicts the given condition that d is positive. Therefore, $a \neq b$, which implies $a < b$.

Because the statement and the converse are both true, we can conclude that for some positive number d , $a < b$ if and only if $b = a + d$. $\square$

Exercise 2.2.4. *Justify the three lemmas marked with (why?) in Proposition 2.2.13 (trichotomy of order for natural numbers).*

Solution.

(a) For all b , $0 \leq b$.

From the definition of addition, $0 + b = b$. By Definition 2.2.11, this implies $0 \leq b$. $\square$

(b) If $a > b$, then $a++ > b$.

From Definition 2.2.11, $a = b + m$, where m is a natural number. Thus, by the definition of addition, $a++ = (b + m)++ = b + m++$. Because $m++$ is positive by Proposition 2.2.8, $a++ > b$. $\square$

(c) If $a = b$, then $a++ > b$.

From Axiom 2.4, if $a = b$, then $a++ = b++$. Recall that $b++ = b + 1$, thus $a++ = b + 1$. By Definition 2.2.11, this implies $a++ > b$. $\square$

Exercise 2.2.5. *Prove Proposition 2.14 (Strong principle of induction): Let m_0 be a natural number, and let $P(m)$ be a property pertaining to an arbitrary natural number m . Suppose that for each $m \geq m_0$, we have the following implication: if $P(m')$ is true for all natural numbers $m_0 \leq m' < m$, then $P(m)$ is also true. Then we can conclude that $P(m)$ is true for all natural numbers $m \geq m_0$.*

Solution. Let $Q(n)$ be the property such that $P(m)$ is true for all $m_0 \leq m < n$. Note that the supposed implication becomes “If $Q(n)$ is true, then $P(n)$ is also true. We induct on n .

First, consider the base case $n \leq m_0$. In this case, $Q(n)$ is vacuously true as there is no such natural number m . Hence, the base case is proven.

Now, we assume inductively that $Q(n)$ is true; we will now prove $Q(n++)$ is true. By the inductive hypothesis, $P(m)$ is true for all m where $m_0 \leq m < n$. According to the supposed implication, because $Q(n)$ is true, $P(n)$ is also true. Therefore, $P(m)$ is true for $m_0 \leq m \leq n$, which implies $Q(n++)$ is also true. This closes the induction. $\square$

Exercise 2.2.6. *Let n be a natural number, and let $P(m)$ be a property pertaining to the natural numbers such that whenever $P(m++)$ is true, then $P(m)$ is true. Suppose that $P(n)$ is also true. Prove that $P(m)$ is true for all natural numbers $m \leq n$; this is known as the principle of backwards induction.*

Solution. Write $Q(n)$ as the property “If $P(n)$ is true, $P(m)$ is true for all natural numbers $m \leq n$. We apply induction on n .

Consider the base case $n = 0$. The statement $m \leq n$ implies $0 = m + x$ for a natural number x . According to Corollary 2.2.9, this leads to $m = x = 0$. Thus, $Q(n)$ is equivalent to “If $P(n)$ is true, $P(n)$ is true.”, which is always true regardless of whether $P(n)$ is true or false. Hence, the base case is proven.

Suppose inductively that $Q(n)$ is true; we now need to show $Q(n++)$ is true. From the definition of $P(m)$, $P(n++)$ being true implies $P(n)$ is true. By the inductive statement, $P(m)$ is true for all natural numbers $m \leq n$. Thus if $P(n++)$ is true, $P(m)$ is true for all $m \leq n++$. This is precisely the definition of $Q(n++)$, so $Q(n++)$ is also true. This closes the induction. $\square$

2.3 Multiplication

Exercise 2.3.1. *Prove Lemma 2.3.2 (Commutativity of multiplication): Let n, m be natural numbers. Then $n \times m = m \times n$.*

Solution. First, we introduce a few lemmas. For all lemmas, we induct on n , keeping m fixed (when it appears).

Lemma. *For any natural number n , $n \times 0 = 0$.*

Proof. In the base case $n = 0$, $0 \times 0 = 0$ is true by the definition of multiplication, and 0 is a natural number. Now suppose inductively that $n \times 0 = 0$; we will now prove $(n++) \times 0 = 0$. By the definition of multiplication, $(n++) \times 0 = (n \times 0) + 0 = 0 + 0 = 0$. $\square$

Lemma. *For any natural numbers n and m , $n \times (m++) = (n \times m) + n$.*

Proof. In the base case $n = 0$, $0 \times m++ = 0$ by the definition of multiplication. Also, $(0 \times m) + 0 = 0$. Thus, the left-hand side is true.

Now suppose inductively that $n \times (m++) = (n \times m) + n$; we must now show $(n++) \times (m++) = ((n++) \times m) + (n++)$. We start on the left-hand side.

$$\begin{aligned}
 & (n++) \times (m++) \\
 &= n \times (m++) + m++ && \text{(Definition 2.3.1)} \\
 &= (n \times m) + n + m++ && \text{(Inductive hypothesis)} \\
 &= (n \times m) + m + n++ && \text{(Addition properties)} \\
 &= ((n++) \times m) + (n++) && \text{(Definition 2.3.1)}
 \end{aligned}$$

This closes the induction. $\square$

We can now prove the main statement $n \times m = m \times n$. Consider the base case $n = 0$. Thus, $0 \times m = m \times 0 = 0$. Hence, the base case is true.

Now suppose inductively that $n \times m = m \times n$; we will now prove $(n++) \times m = m \times (n++)$. By the definition of multiplication, $(n++) \times m = (n \times m) + m$. From the second lemma, the right-hand side is also equivalent to $(n \times m) + m$. Because both sides are equal, the statement is true. This closes the induction. $\square$

Exercise 2.3.2. *Prove Lemma 2.3.3 (Positive natural numbers have no zero divisors): Let m, n be natural numbers. Then $n \times m = 0$ if and only if at least one of n, m is equal to zero. In particular, if n and m are both positive, then nm is also positive.*

Solution. We will prove the second statement first. For the sake of contradiction, suppose there exist positive numbers m, n such that $mn = 0$. Because m, n are positive numbers, write $m = p++$, $n = q++$, thus $(q++) \times (p++) = 0$. From the definition of multiplication, $(q \times (p++)) + (p++) = 0$. According to Corollary 2.2.9, this implies $q \times (p++) = p++ = 0$, which is contradictory as Axiom 2.3 states 0 is not the successor of any natural number. Hence, if m, n are both positive, then mn is positive.

Now consider the first statement. The left-to-right implication is simply the contrapositive of the second statement, which is true. Observe the right-to-left implication. If $n = 0$, then $0 \times m = 0$ by the definition of multiplication. If $m = 0$, then by commutativity of multiplication, $m \times 0 = 0 \times m = 0$. Hence, the statements are proven as desired. $\square$

Exercise 2.3.3. *Prove Proposition 2.3.5 (Associativity of multiplication): For any natural numbers a, b, c , we have $(a \times b) \times c = a \times (b \times c)$.*

Solution. We induct on a , keeping b and c fixed.

First, consider the base case $a = 0$. According to the definition of multiplication, the left-hand side leads to $(0 \times b) \times c = 0 \times c = 0$. Similarly, the right-hand side results in $0 \times (b \times c) = 0$. Because both sides are equal, our base case is proven.

Now we assume inductively that $(a \times b) \times c = a \times (b \times c)$; we need to show that $((a + 1) \times b) \times c = (a + 1) \times (b \times c)$. By the distributive law, the left-hand side results in $((a + 1) \times b) \times c = (a \times b + b) \times c = (a \times b) \times c + b \times c$. The right-hand side produces $(a + 1) \times (b \times c) = a \times (b \times c) + b \times c$, which, by the inductive hypothesis, is equal to $(a \times b) \times c + b \times c$. Thus, both sides are equal, and this closes the induction. $\square$

Exercise 2.3.4. *Prove the identity $(a+b)^2 = a^2 + 2ab + b^2$ for all natural numbers a, b .*

Solution. By Proposition 2.3.4 (distributive law) and Proposition 2.3.5 (commuta-

tivity of multiplication):

$$\begin{aligned}(a + b)^2 &= (a + b)(a + b) \\ &= a(a + b) + b(a + b) \\ &= a^2 + ab + ab + b^2 \\ &= a^2 + 2ab + b^2\end{aligned}$$

Exercise 2.3.5. *Prove Proposition 2.3.9 (Euclidean algorithm): Let n be a natural number, and let q be a positive number. Then there exists natural numbers m, r such that $0 \leq r < q$ and $n = mq + r$.*

Solution. We induct on n , fixing q .

First, observe the base case $n = 0$. There is an obvious solution of $m = 0, r = 0$. Hence, the base case is proven.

Now, suppose inductively that there exist natural numbers m, r such that $0 \leq r < q$ and $n = mq + r$; we now have to show there exists natural numbers m', r' such that $0 \leq r' < q$ and $n + 1 = m'q + r'$. By the induction hypothesis, $n + 1 = mq + r + 1$. Since $r < q$, $r + 1 \leq q$. Thus, we have two cases:

- (a) $r + 1 < q$: In this case, we can have $n + 1 = mq + (r + 1)$ with $0 \leq r + 1 < q$. Thus, we can have $m' = m, r' = r + 1$.
- (b) $r + 1 = q$: In this case, we have $n + 1 = mq + q = (m + 1)q$ by the distributive law. Therefore, $m' = m + 1, r' = 0$.

This closes the induction.